Project Development Phase
Model Performance Test

Date	15 April 2026
Team ID	---------
Project Name	Project – Online Fraud Detection System
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot

1.	Metrics	**Classification Model:** Confusion Matrix – Accuray Score- Classification Report -	1.RandomForest clasifier [ ] rfc=RandomForestClassifier() rfc.fit(x_train, y_train) y_test_predict1=rfc.predict(x_test) test_accuracy=accuracy_score(y_test,y_test_predict1) test_accuracy 0.9997004661547614 [ ] y_train_predict1=rfc.predict(x_train) train_accuracy=accuracy_score(y_train,y_train_predict1) train_accuracy 1.0 [ ] pd.crosstab(y_test,y_test_predict1) col_0 is Fraud is not Fraud isFraud is Fraud 807 336 is not Fraud 23 1197363 [ ] print(classification_report(y_test,y_test_predict1)) precision recall f1-score support is Fraud 0.97 0.71 0.82 1143 is not Fraud 1.00 1.00 1.00 1197386 accuracy 1.00 1198529 macro avg 0.99 0.85 0.91 1198529 weighted avg 1.00 1.00 1.00 1198529 2.Decision Tree classifier [ ] from sklearn.tree import DecisionTreeClassifier dtc=DecisionTreeClassifier() dtc.fit(x_train, y_train) y_test_predict2=dtc.predict(x_test) test_accuracy=accuracy_score(y_test,y_test_predict2) test_accuracy 0.9996912882374978 [ ] y_train_predict2=dtc.predict(x_train) train_accuracy=accuracy_score(y_train,y_train_predict2) train_accuracy 1.0 [ ] pd.crosstab(y_test,y_test_predict2) col_0 is Fraud is not Fraud isFraud is Fraud 1193 245 is not Fraud 204 1496519 [ ] print(classification_report(y_test,y_test_predict2)) precision recall f1-score support is Fraud 0.85 0.83 0.84 1438 is not Fraud 1.00 1.00 1.00 1496723 accuracy 1.00 1498161 macro avg 0.93 0.91 0.92 1498161 weighted avg 1.00 1.00 1.00 1498161
			4 Xgboost Classifier [ ] import xgboost as xgb xgb1 = xgb.XGBClassifier() xgb1.fit(x_train,y_train1) y_test_predict5=xgb1.predict(x_test) test_accuracy=accuracy_score(y_test1,y_test_predict5) test_accuracy 0.9997904401688998 [ ] y_train_predict5=xgb1.predict(x_train) train_accuracy=accuracy_score(y_train1,y_train_predict5) train_accuracy 0.9998602933377643 [ ] pd.crosstab(y_test1,y_test_predict5) col_0 0 1 row_0 0 642 172 1 32 972623 [ ] print(classification_report(y_test1,y_test_predict5)) precision recall f1-score support 0 0.95 0.79 0.86 814 1 1.00 1.00 1.00 972655 accuracy 1.00 973469 macro avg 0.98 0.89 0.93 973469 weighted avg 1.00 1.00 1.00 973469

2.	Tune the Model	Hyperparameter Tuning -	The accuracy for the model is high without hyperparameter tunning and the type 2 error is also very low.